CMSC 420: AVL Trees

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1 Problems **2**

1 Problems

1. Consider the following AVL tree:

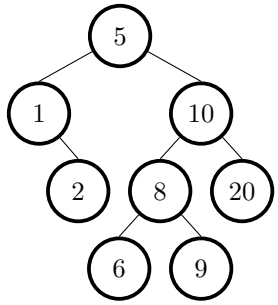

Figure 1: An AVL Tree

The following questions are non-cumulative:

(a) Suppose we insert a key x and the tree doesn't need rebalancing. Assuming x is an integer and all the keys are different, what can we say about x? You may use specific values or inequalities to express your answer.

What is true about x?	

(b) Suppose we delete a key x, using inorder predecessors when replacements are needed, and the tree doesn't need rebalancing. What can we say about x? You may use specific values or inequalities to express your answer.

What is true about x?	

(c) Suppose we delete the key $x = 2$. How many single rotations will be necessary to rebalance the tree? Note that a left-right rotation would count as two "single rotations".

How many single rotations?	

2. Here is a diagram of the right rotation of a node:

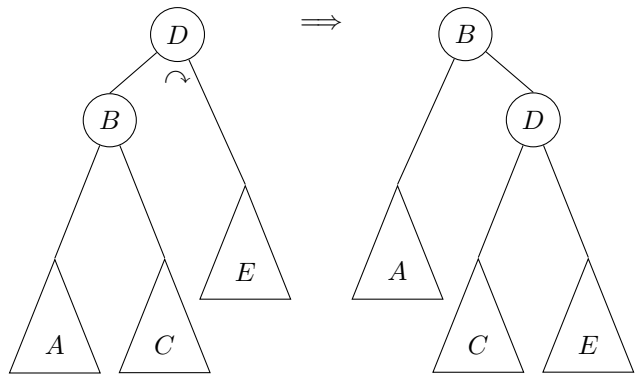

Figure 2: Right Rotation at a Node

Suppose that nodes B and D each contain values `B.leftheight`, `B.rightheight`, `D.leftheight`, and `D.rightheight` which contain the height of their respective subtrees?

(a) After the rotation what would `B.leftheight` be in terms of the values before the rotation?

(b) After the rotation what would `B.rightheight` be in terms of the values before the rotation?

(c) After the rotation what would `D.leftheight` be in terms of the values before the rotation?

(d) After the rotation what would `D.rightheight` be in terms of the values before the rotation?

3. In class we drew pictures of the effect of a right rotation on a left-left heavy AVL tree after insertion and the effect of a left-right rotation on a left-right heavy AVL tree after insertion. Draw appropriate pictures of the effect of a left rotation on a right-right heavy AVL tree after insertion and the effect of a right-left rotation on a right-left heavy AVL tree after insertion.

4. Starting with the following AVL tree:

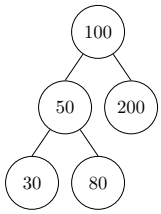

Figure 3: An AVL Tree

Perform the following operations cumulatively in the order given. For each operation draw:

- The tree after the operation but before any required rebalancing.
- The tree after each step of any required rebalancing operations. For example if a single rotation is required, just draw the result, but if a double-rotation is required draw the result after each component rotation.

(a) Insert 90.

(b) Insert 10.

5. Given the following AVL tree:

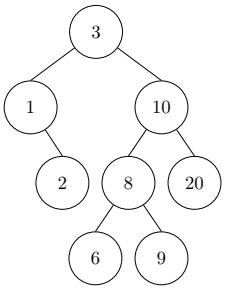

Figure 4: An AVL Tree

For the following questions start with the above tree. In other words the questions do not build from one another!

(a) Show the process (each step) through which the key 5 is inserted into the tree.

(b) Show the process (each step) through which the key 2 is deleted from the tree.

6. In class we showed that for an AVL tree with height h and n nodes that when h is even then $N(h) > 2N(h-2)$ implies that $h = \mathcal{O}(\lg n)$, where $N(h)$ is the minimum number of nodes in an AVL tree of height h. Prove that the statement is also true when h is odd.

7. Here is the code (in Python) for the left rotation in an AVL tree:

```
def rotateleft(root: Node) -> Node:
    RightChild = root.rightchild
    RightLeftSubtree = RightChild.leftchild
    RightChild.leftchild = root
    root.rightchild = RightLeftSubtree
return RightChild
```

Suppose that the `Node` class also contains variables `Node.leftheight` and `Node.rightheight` which store the heights of the left and right subtrees.

Write down the two (and only two!) necessary lines which will go immediately before the `return` statement so that these two variables are updated wherever necessary. Your result does not need to be proper Python, pseudocode will suffice.

8. Suppose b_h equals the number of different AVL tree structures of height h. We don't care about the contents, just the overall structure.

   (a) Calculate b_0 and b_1 via illustrations.

   (b) Prove that for $h \geq 2$ we have:

   $$b_h = 2b_{h-1}b_{h-2} + b_{h-1}^2$$

   (c) Use this to calculate b_h for $h = 2, 3, 4$.

   (d) Verify your calculation for $h = 2$ by illustrating all possible structures.

9. The following statement is either true or false. State which you believe it to be and justify. Note: The depth of a node is the distance (number of edges) from the node to the root.

   Statement: For any AVL tree if L_i and L_j are leaves, then:

$$\text{depth}(L_i) - \text{depth}(L_j) \in \{-1, 0, 1\}$$

10. Define an AVL-2 tree to be a BST where each node v satisfies $\text{balance}(v) \in \{-2, -1, 0, 1, 2\}$. Let $N(h)$ be the minimum number of nodes in an AVL-2 tree of height h. Prove that $h = \mathcal{O}(\lg n)$ for the case where h is a multiple of 3.

11. Suppose an AVL tree with height h has n nodes. Let F_i be the ith Fibonacci number, so $F_0 = 0$, $F_1 = 1$, and $F_j = F_{j-1} + F_{j-2}$ for $j \geq 2$. Prove that:

$$n \geq F_{h+3} - 1$$

12. Prove that an AVL tree with n nodes has height $\Omega(\lg n)$.

13. Prove by induction that for any integer $\delta \geq 1$ it's possible to construct an AVL tree containing two leaves whose depth differs by δ.

14. For some fixed but unknown positive integer k define an AVL-k tree to be a BST where each node v satisfies:

$$\text{balance}(v) \in \{-k, ..., -2, -1, 0, 1, 2, ..., k\}$$

Let $N(h)$ be the minimum number of nodes possible in an AVL-k tree of height h. Use this to prove that $h = \mathcal{O}(\lg n)$. For simplicity of some calculations you can assume that $h \geq k$.

15. Consider the following AVL tree:

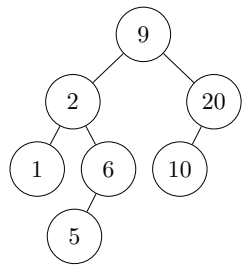

Figure 5: An AVL Tree

The following questions are non-cumulative. For each question assume x is a positive integer and all the keys are different.

Please read these questions carefully!

(a) List all the keys x with balance$(x) = \pm 1$. List in increasing order.

Possible Keys	

(b) Suppose we insert a key x and the tree does not need rebalancing. What are the possible values of x? You may use specific values or inequalities to express your answer.

Possible Keys	

(c) Suppose we delete a key x, using inorder predecessors when replacements are needed, and the tree does need rebalancing. What are the possible values of x? List in increasing order.

Possible Keys	

(d) Suppose we delete a key x, using inorder successors when replacements are needed, and the tree does need rebalancing. What are the possible values of x? List in increasing order.

Possible Keys	

(e) Suppose we delete the key $x = 20$. How many single rotations will be necessary to rebalance the tree? Note that a LR or RL rotation would count as two "single rotations". [5 pts]

How Many?	

16. If a binary tree has the structure shown:

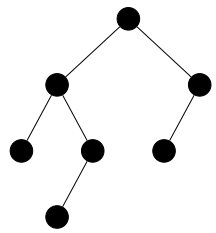

Figure 6: A Binary Tree

If we perform a right rotation at the root, draw the resulting tree structure.

17. Suppose `x` is the root of a nonempty binary tree. Suppose we run this algorithm:

```
x = root
while x != None:
    while x.left != None:
        rightrotation at x
        x = x.parent
    end while
    x = x.right
end while
```

What will the binary tree look like after this is run? A single sentence will suffice.

18. Here is a diagram of the right rotation of a node:

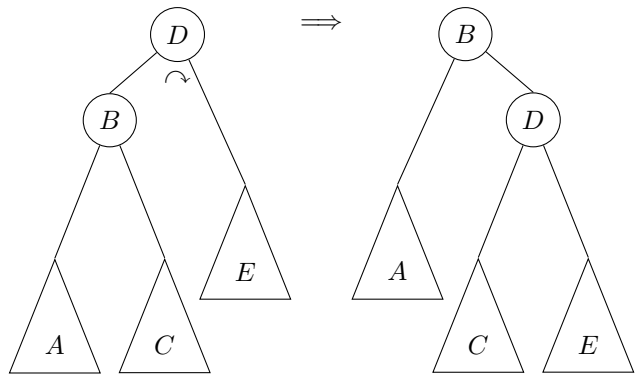

Figure 7: Right Rotation at a Node

Suppose we have the values `B.leftheight`, `B.rightheight`, `D.leftheight`, and `D.rightheight` which store the heights of the subtrees. Out of these four only `D.leftheight` and `B.rightheight` change with the rotation. What would the new values be in terms of the old values?

`D.leftheight =`	
`B.rightheight =`	

19. Define $N(h)$ to be the minimum number of nodes possible in an AVL tree with height h.
    We know from class that $N(h) > 2N(h-2)$.
    Use digging down to prove that $h = \mathcal{O}(\lg n)$ when h is odd.